

* Circular Convolution (for periodic DTS)

$$x[n] = \boxed{1, 2, 3, 5}, 1, 2, 3, 5$$

$$L_1 = 4 \text{ (length)}$$

$$h[n] = \boxed{4, 5, 6, 7}, 4, 5, 6, 7$$

$$L_2 = 4$$

$$Q. \quad x[n] = \boxed{1, 2, 3, 4}, \boxed{1, 2, 3, 4}, \boxed{1, 2, 3, 4}$$

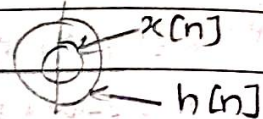
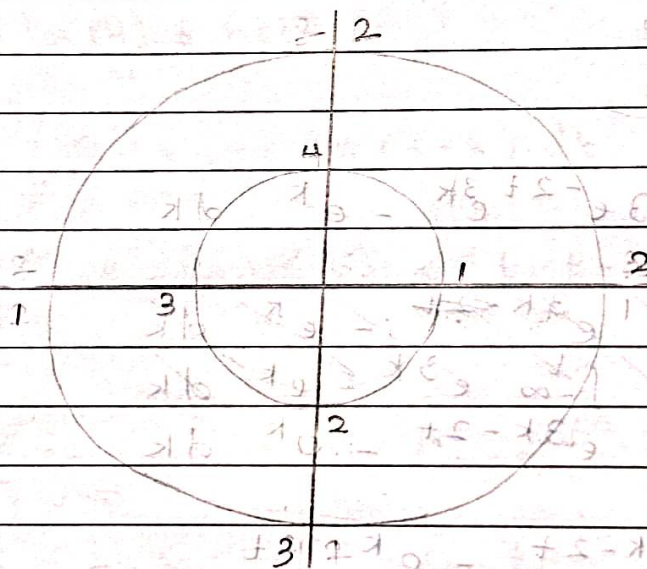
$$L_1 = 4$$

$$h[n] = \boxed{2, 1, 3, 2}, \boxed{2, 1, 3, 2}$$

$$L_2 = 4$$

Find $y[n]$

$$y[n] = x[n] \circledast h[n]$$

divide in 4 parts becaus $L=4$ 

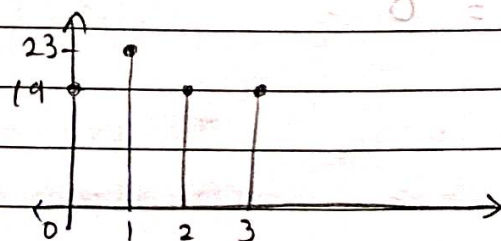
$$y[0] = 1 \times 2 + 2 \times 2 + 3 \times 3 + 4 \times 1 = 2 + 4 + 9 + 4 = 19$$

$$y[1] = 1 \times 1 + 2 \times 2 + 3 \times 2 + 3 \times 4 = 1 + 4 + 6 + 12 = 23$$

$$y[2] = 1 \times 3 + 2 \times 1 + 3 \times 2 + 2 \times 4 = 3 + 2 + 6 + 8 = 19$$

$$y[3] = 1 \times 2 + 2 \times 3 + 3 \times 1 + 4 \times 2 = 2 + 6 + 3 + 8 = 19$$

$$y[n] = 19, 23, 19, 19$$



$$\begin{bmatrix} 2 & 2 & 3 & 1 \\ 1 & 2 & 2 & 3 \\ 3 & 1 & 2 & 2 \\ 2 & 3 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 19 \\ 23 \\ 19 \\ 19 \end{bmatrix}$$

Q. $x[n] = 1, 2, 3, 1, 2, 3$ $L_1 = 3$

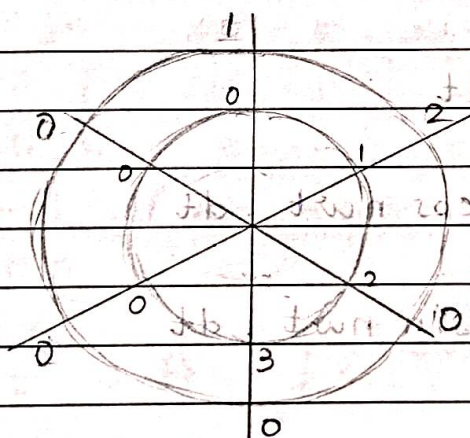
$h[n] = 2, 1, 2, 1$ $L_2 = 2$

Find $y[n]$.

$\text{LCM}(2, 3) = 6$

$x[n] = 1, 2, 3, 0, 0, 0$

$h[n] = 2, 1, 0, 0, 0, 0$



$y[0] = 1 \times 2 + 2 \times 1 + 3 \times 0 = 2 + 2 = 4$

$y[1] = 2 \times 2 + 3 \times 1 = 4 + 3 = 7$

$y[2] = 3 \times 2 = 6$

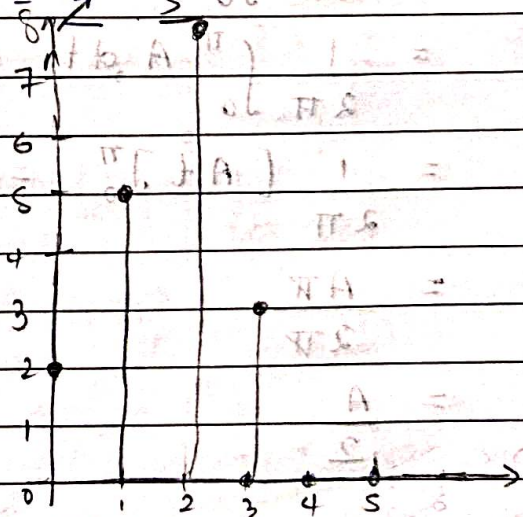
$y[3] = 0$

$y[4] = 0$

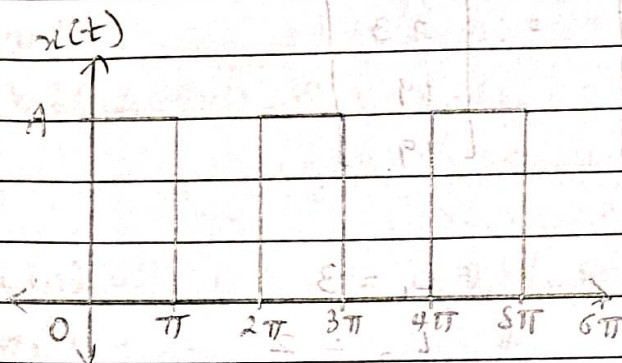
$y[5] = 1 \times 1 = 1$

$y[6] = 0$

$y[n] = 4, 7, 6, 0, 0, 1$



Fourier Series



$T = 2\pi$... time period of the periodic wave form.

$$\omega = 2\pi f = 2\pi \frac{1}{T} = 1 \quad \dots \quad \omega = \text{omega}$$

$f = \text{frequency}$

$f = 1/T$

$$a_0 = \frac{1}{T} \int_0^T x(t) \cdot dt$$

$$a_n = \frac{2}{T} \int_0^T x(t) \cdot \cos n\omega t \cdot dt$$

$$b_n = \frac{2}{T} \int_0^T x(t) \cdot \sin n\omega t \cdot dt$$

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} x(t) \cdot dt$$

$$= \frac{1}{2\pi} \int_0^{\pi} A \cdot dt + \frac{1}{2\pi} \int_{\pi}^{2\pi} 0 \cdot dt$$

$$= \frac{1}{2\pi} [At]_0^{\pi} = \frac{A}{2\pi} [\pi - 0]$$

$$= \frac{A\pi}{2\pi}$$

$$= \frac{A}{2}$$

$$\begin{aligned}
 a_n &= \frac{2}{2\pi} \int_0^{2\pi} x(t) \cdot \cos n\omega t \, dt \\
 &= \frac{1}{\pi} \int_0^{\pi} A \cdot \cos n\omega t \, dt \\
 &= \frac{A}{\pi} \left[\frac{\sin n\omega t}{n\omega} \right]_0^{\pi} \\
 &= \frac{A}{\pi n\omega} [\sin n\omega\pi - \sin 0] \\
 &= \frac{A \cdot \sin n\pi}{n\pi} = \underline{0} \quad \dots \omega = 1/T
 \end{aligned}$$

$$\begin{aligned}
 b_n &= \frac{2}{2\pi} \int_0^{2\pi} x(t) \cdot \sin n\omega t \, dt \\
 &= \frac{1}{\pi} \int_0^{\pi} A \cdot \sin n\omega t \, dt \\
 &= \frac{A}{\pi} \left[-\frac{\cos n\omega t}{n\omega} \right]_0^{\pi} \\
 &= \frac{-A}{\pi n} [\cos n\pi - \cos 0] \\
 &= \frac{-A}{\pi n} [(-1)^n - 1]
 \end{aligned}$$

For $n = \text{even}$; $\frac{-A}{\pi n} [0] = 0$

For $n = \text{odd}$;

$$\frac{2A}{\pi n}$$

Q. Find FS for..

 $x(t)$ A A 0 π 2π 3π

$$T = 2\pi$$

0

$$\omega = 2\pi f = 2\pi \cdot \frac{1}{T} = \frac{2\pi}{2\pi} = 1$$

$$x(t) = \frac{A}{\pi} t$$

$$0 < t < \pi$$

$$= 0$$

$$\pi < t < 2\pi$$

Using slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$

$$\therefore m = \frac{A - 0}{\pi - 0} = \frac{A}{\pi}$$

since $y = mx$,

$$x(t) = \frac{A}{\pi} t$$

$$q_0 = \frac{1}{T} \int_0^T x(t) \cdot dt$$

$$= \frac{1}{2\pi} \int_0^{2\pi} x(t) dt$$

$$= \frac{1}{2\pi} \int_0^{\pi} \frac{A}{\pi} t \cdot dt + \frac{1}{2\pi} \int_{\pi}^{2\pi} 0 \cdot dt$$

$$= \frac{A}{2\pi^2} \int_0^{\pi} t \cdot dt$$

$$= \frac{A}{2\pi^2} \left[\frac{t^2}{2} \right]_0^\pi = \frac{A}{4\pi^2} [\pi^2 - 0]$$

$$= \frac{A}{4}$$

$$a_n = \frac{2}{T} \int_0^T x(t) \cos(n\omega t) dt$$

$$= \frac{2}{2\pi} \int_0^\pi \frac{At}{\pi} \cos(n\omega t) dt \quad A = \dots \omega = 1$$

$$= \frac{A}{\pi^2} \int_0^\pi t \cos(nt) dt$$

$$= \frac{A}{\pi^2} \left[\frac{-t \sin(nt)}{n} + \frac{(1 - \cos nt)}{n^2} \right]_0^\pi$$

$$= \frac{A}{n\pi^2} \left[-t \sin(nt) + \frac{\cos(nt)}{n} \right]_0^\pi$$

$$= \frac{A}{n\pi^2} \left[-\pi \sin(n\pi) + \frac{\cos(n\pi)}{n} - 0 \sin(0) - \frac{\cos(0)}{n} \right]$$

$$= \frac{A}{n\pi^2} \left[\frac{\cos(n\pi)}{n} - \frac{1}{n} \right]$$

$$= \frac{A}{n\pi^2} \left[\frac{(-1)^n - 1}{n} \right]$$

$$= \frac{A}{n^2\pi^2} [(-1)^n - 1]$$

For $n = \text{even}$; $a_n = \frac{A}{n^2\pi^2} [1 - 1] = 0$

$n = \text{odd}$; $a_n = \frac{A}{n^2\pi^2} [-1 - 1]$

$$= \frac{-2A}{n^2\pi^2}$$

$$\sin(n\pi) = 0 \quad \sin(2n\pi) = 0$$

$$\cos(n\pi) = (-1)^n \quad \cos(2n\pi) = 1$$

$$b_n = \frac{2}{T} \int_0^T x(t) \cdot \sin(n\omega t) dt$$

$$= \frac{2}{2\pi} \int_0^\pi \frac{At}{\pi} \sin(nt) dt \quad \dots \omega = 1$$

$$= \frac{A}{\pi^2} \int_0^\pi t \sin(nt) dt$$

$$= \frac{A}{\pi^2} \left[\frac{t \cdot \cos(nt)}{n} - \left(\frac{1}{n^2} - \frac{\sin(nt)}{n^2} \right) \right]_0^\pi$$

$$= \frac{A}{\pi^2} \left[\frac{t \cdot \cos(nt)}{n} + \frac{\sin(nt)}{n^2} \right]_0^\pi$$

$$= \frac{A}{\pi^2} \left[\frac{\pi \cdot \cos(n\pi)}{n} + \frac{\sin(n\pi)}{n^2} \right]$$

$$= \frac{A}{\pi^2} \left[\frac{\pi \cdot (-1)^n}{n} - \frac{\sin(0)}{n^2} \right]$$

$$= \frac{A}{\pi^2} \left[\frac{\pi \cdot (-1)^n}{n} \right]$$

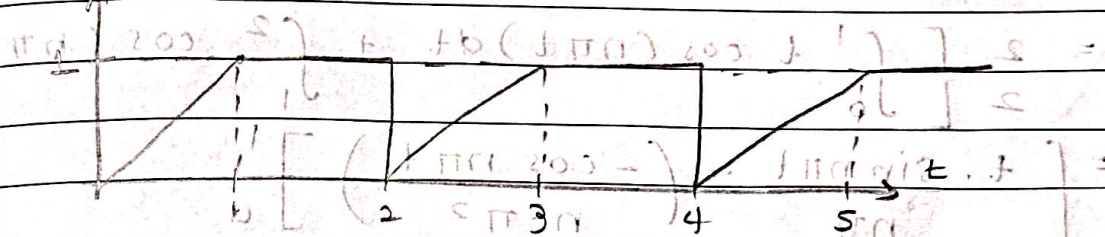
$$= \frac{A}{n\pi} [(-1)^n]$$

$$\text{For } n = \text{even}; \quad b_n = \frac{A}{n\pi}$$

$$\text{For } n = \text{odd}; \quad b_n = \frac{-A}{n\pi}$$

Q. Find FS for : $x(t)$ as shown below

$x(t)$



$$T = 2$$

$$\omega = 2\pi f = 2\pi \frac{1}{T} = \frac{2\pi}{2} = \pi \text{ rad/s}$$

$$x(t) = \begin{cases} t & 0 \leq t < 1 \\ 1 & 1 \leq t < 2 \\ 0 & \text{elsewhere} \end{cases}$$

$$a_0 = \frac{1}{T} \int_0^T x(t) dt = \frac{1}{2} \left[\int_0^1 t dt + \int_1^2 1 dt \right]$$

$$= \frac{1}{2} \left[\left[\frac{t^2}{2} \right]_0^1 + \left[t \right]_1^2 \right]$$

$$= \frac{1}{2} \left[\frac{1}{2} + 1 \right]$$

$$= \frac{1}{2} \left[\frac{3}{2} \right]$$

$$= \frac{3}{4}$$

$$a_n = \frac{1}{T} \int_0^T x(t) \cos(n\omega t) dt = \frac{1}{2} \left[\int_0^1 t \cos(n\pi t) dt + \int_1^2 \cos(n\pi t) dt \right]$$

$$= \frac{1}{2} \left[\left[\frac{t \sin(n\pi t)}{n\pi} - \frac{\cos(n\pi t)}{n^2\pi^2} \right]_0^1 + \left[\frac{\sin(n\pi t)}{n\pi} \right]_1^2 \right]$$

$$= \frac{1}{2} \left[\left(\frac{1 \sin(n\pi)}{n\pi} - \frac{\cos(n\pi)}{n^2\pi^2} \right) - \left(\frac{0 \sin(0)}{n\pi} - \frac{\cos(0)}{n^2\pi^2} \right) + \left(\frac{\sin(2n\pi)}{n\pi} - \frac{\sin(n\pi)}{n\pi} \right) \right]$$

$$= \frac{1}{2} \left[\left(\frac{0}{n\pi} - \frac{\cos(n\pi)}{n^2\pi^2} \right) - \left(\frac{0}{n\pi} - \frac{1}{n^2\pi^2} \right) + \left(\frac{0}{n\pi} - \frac{0}{n\pi} \right) \right]$$

$$\begin{aligned}
 a_n &= \frac{2}{T} \int_0^T x(t) \cdot \cos(n\omega t) dt \\
 &= \frac{2}{2} \left[\int_0^1 t \cos(n\pi t) dt + \int_1^2 \cos(n\pi t) dt \right] \\
 &= \left[\frac{t \cdot \sin n\pi t}{n\pi} - \left(\frac{-\cos n\pi t}{n^2\pi^2} \right) \right]_0^1 \\
 &\quad + \left[\frac{\sin n\pi t}{n\pi} \right]_1^2 \\
 &= \left[\frac{1 \cdot \sin(n\pi)}{n\pi} + \frac{\cos(n\pi)}{n^2\pi^2} - \frac{0 \cdot \cos(0)}{n^2\pi^2} \right] \\
 &\quad + \left[\frac{\sin(2n\pi)}{n\pi} - \frac{\sin(n\pi)}{n\pi} \right] \\
 &= \left[0 + \frac{(-1)^n}{n^2\pi^2} - 0 - \frac{1}{n^2\pi^2} \right] + [0 - 0] \\
 &= \frac{1}{n^2\pi^2} [(-1)^n - 1]
 \end{aligned}$$

For $n = \text{even}$; $a_n = \frac{1}{n^2\pi^2} [1 - 1] = 0$

$n = \text{odd}$; $a_n = \frac{1}{n^2\pi^2} [-1 - 1] = -\frac{2}{n^2\pi^2}$

$= -\frac{2}{n^2\pi^2}$

$n^2\pi^2$

$$b_n = \frac{2}{T} \int_0^T x(t) \cdot \sin(n\omega t) dt$$

$$= \frac{2}{2} \left[\int_0^1 t \sin(n\pi t) dt + \int_1^2 \sin(n\pi t) dt \right]$$

$$= \left[\frac{-t \cdot \cos n\pi t}{n\pi} - \left(\frac{-\sin n\pi t}{n^2\pi^2} \right) \right]_0^1$$

$$+ \left[\frac{-\cos n\pi t}{n\pi} \right]_1^2$$

$$\begin{aligned}
 &= \left[\frac{-t \cos n\pi t}{n\pi} + \frac{\sin n\pi t}{n^2\pi^2} \right]_0^1 + \left[\frac{-\cos n\pi t}{n\pi} \right]_1^2 \\
 &= \left[\frac{-\cos n\pi}{n\pi} + \frac{\sin n\pi}{n^2\pi^2} + 0 - \frac{\sin(0)}{n^2\pi^2} \right] \\
 &\quad + \left[\frac{-\cos 2n\pi}{n\pi} + \frac{\cos n\pi}{n\pi} \right] \\
 &= \frac{-\cos n\pi}{n\pi} - \frac{1}{n\pi} + \frac{\cos n\pi}{n\pi} \\
 &= \frac{-1}{n\pi}
 \end{aligned}$$

After evaluating a_0 , a_n and b_n , the final Fourier series representation is given by

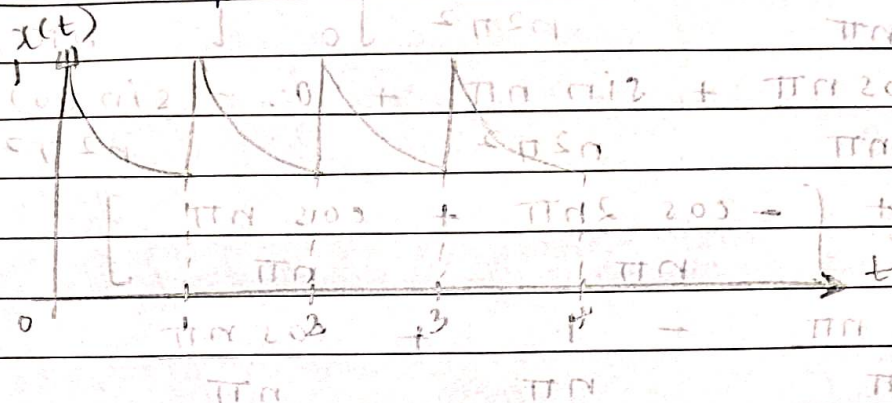
$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega t + b_n \sin n\omega t)$$

* Exponential Fourier Series :

$$x(t) = \sum_{n=-\infty}^{\infty} c_n \cdot e^{jn\omega t}$$

$$\text{where } c_n = \frac{1}{T} \int_0^T x(t) \cdot e^{-jn\omega t} \cdot dt$$

Q. Solve exponential FS for $x(t) = e^{-t}$ for $0 < t < 1$



$$T = 1$$

$$\omega = 2\pi f = \frac{2\pi}{T} = 2\pi$$

$$x(t) = e^{-t} \quad 0 < t < 1$$

$$c_n = \frac{1}{T} \int_0^T x(t) \cdot e^{-jn\omega t} dt = \frac{1}{1} \int_0^1 e^{-t} \cdot e^{-jn2\pi t} dt$$

$$= \frac{1}{1} \int_0^1 e^{-t} \cdot e^{-jn2\pi t} dt$$

$$= \int_0^1 e^{-t} (1 + jn2\pi) dt$$

$$= \left[\frac{e^{-t} (1 + jn2\pi)}{-(1 + jn2\pi)} \right]_0^1$$

$$= \left[\frac{e^{-(1 + jn2\pi)} - 1}{-(1 + jn2\pi)} \right]$$

$$= \frac{1 - e^{-(1 + jn2\pi)}}{1 + jn2\pi}$$

$$= \frac{1}{1 + jn2\pi} [e^{-(1 + jn2\pi)} - 1]$$

$$e^{-1} \cdot e^{-jn2\pi} = -1 \quad [e^{-1} \cdot 1 - 1]$$

$$e^{-jn2\pi} = \cos n2\pi - j \sin n2\pi$$

$$= 1 - 0$$

$$e^{-i\theta} = \cos \theta - i \sin \theta$$